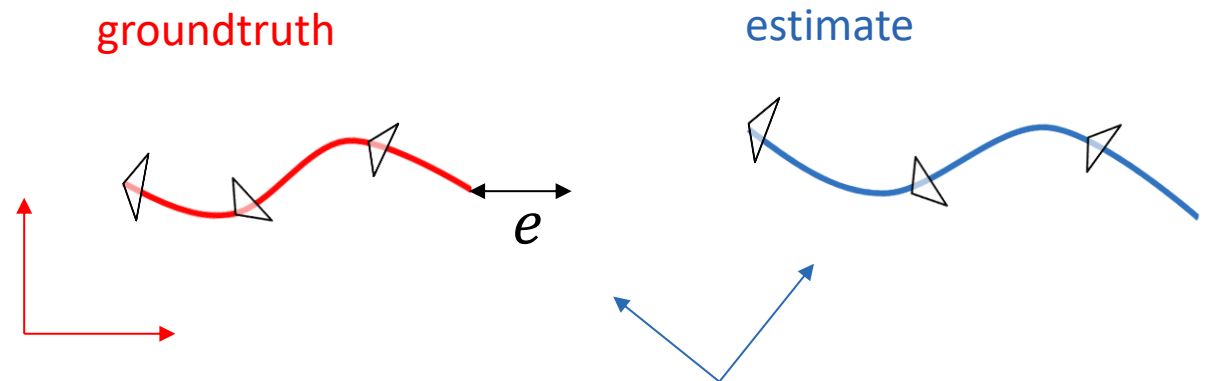


How can we evaluate the accuracy of VO/SLAM algorithms?

- Idea: **compare the estimated trajectory against ground truth** trajectory (from GPS, motion tracking systems), but the key question is **what error metric should be used?**
- Issues:
 - Different reference frames
 - Different scale
- **Naïve solution (not used anymore):** Maybe align the first poses and **measure the end-pose error?**
- **Not repeatable:**
 - Most VOs are **non-deterministic** (e.g., **RANSAC, multithreading**) → every time you run your VO on the same dataset, you get different results
- **Not meaningful:**
 - sensitive to the **trajectory shape** (the number of turns of a trajectory greatly affects the end-pose error)
 - **does not capture the error statistics**



Metric 1: Absolute Trajectory Error (ATE)

- **Step 1:** align the estimated trajectory to the ground truth **from the start to the end** using a similarity transformation (i.e., R, T, s) by minimizing the sum of square position errors

$$R, T, s = \underset{R, T, s}{\operatorname{argmin}} \sum_{k=0}^n \|\hat{C}_k - sRC_k - T\|^2$$

Parameters of the similarity transformation
that we want to find

groundtruth positions

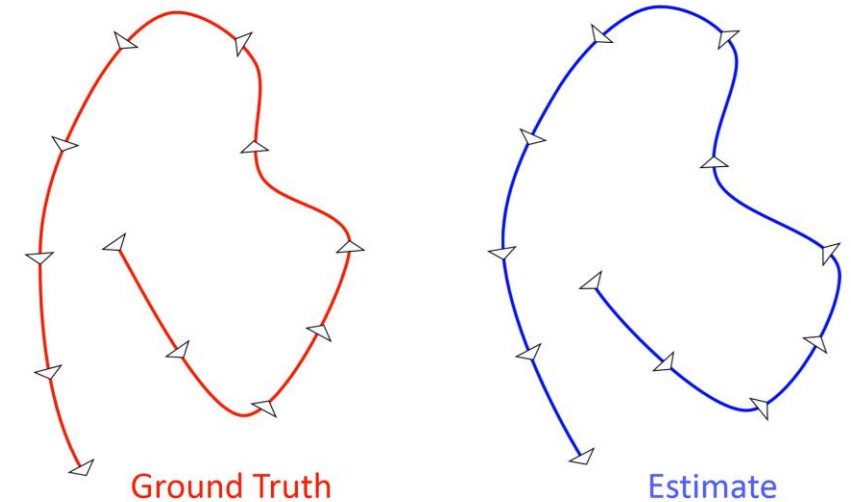
estimated positions

- **Step 2:** compute Root Mean Square Error (RMSE) after alignment:

$$RMSE = \sqrt{\frac{\sum_{k=1}^n \|\hat{C}_k - sRC_k - T\|^2}{n}}$$

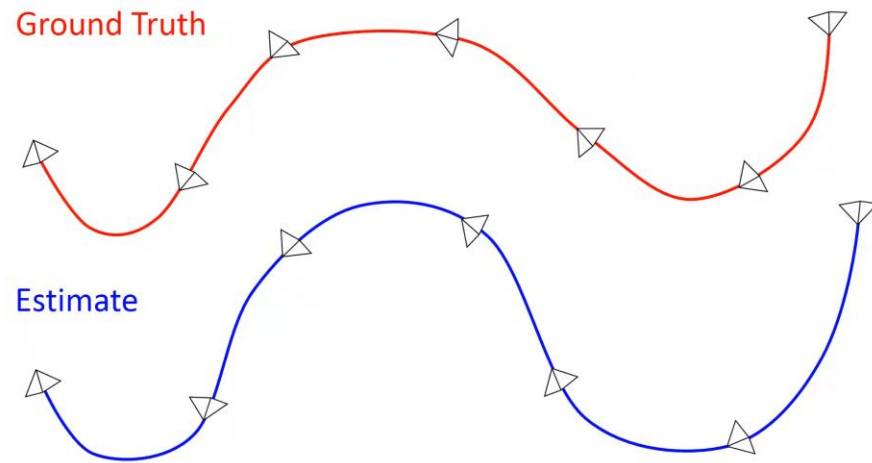
- **Pros and cons:**

- ✓ Single-number metric
- ✓ Captures the global error (accuracy of the global trajectory)
- ✗ Does not capture the relative error (accuracy of the local trajectory estimate)



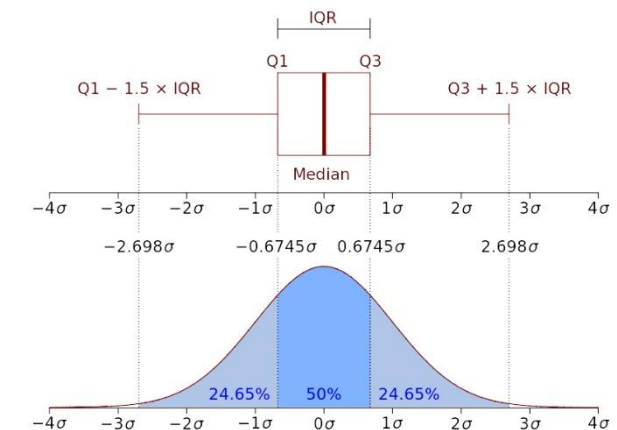
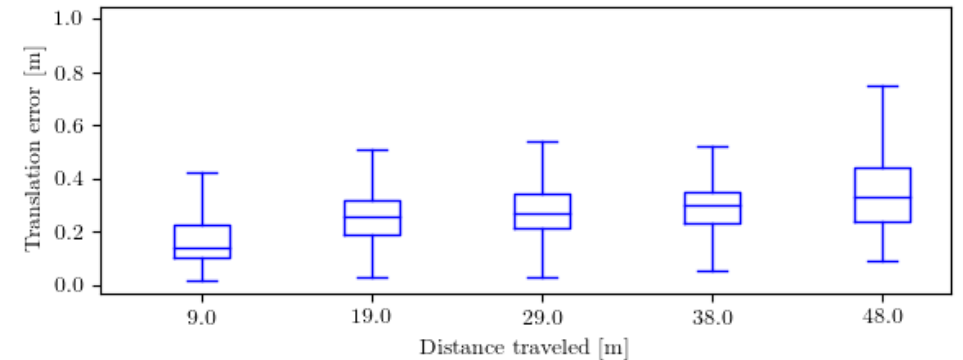
Metric 2: Relative Trajectory Error (RTE)

- Computes **error statistics** of sub-trajectories of specified lengths



- Pros and cons:

- ✓ Informative statistics: captures the relative error (accuracy of the local trajectory estimate)
- ✗ Complicated to compute and rank, but the good news is that there is code for it 😊
(toolbox, link below)



Boxplots are good to visualize error statistics via interquartile ranges ([link](#))

